

TARNAVENI, Romania

WMO: 151650

Lat: 46.3600N

Lon: 24.2261E

Elev: 524

StdP: 95.18

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.8	-9.5	-15.5	1.0	-11.0	-12.6	1.4	-7.7	13.8	1.0	11.8	-0.3	1.7	340	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	29.9	19.6	28.4	19.2	27.0	18.8	21.0	27.4	20.2	26.5	19.5	25.4	2.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.5	23.7	18.2	13.9	23.0	17.5	13.3	22.2	63.3	27.2	60.5	26.4	58.0	25.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	9.9	8.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-2.2	1.4	5.2	10.4	15.4	19.1	20.1	21.0	16.0	10.7	5.5	-0.1
	DBStd	8.93	4.66	4.68	4.66	5.26	3.61	3.43	3.02	3.38	4.58	4.35	4.86	4.07
	HDD10.0	1354	377	243	159	59	4	0	0	0	3	47	148	314
	HDD18.3	3216	635	475	408	239	102	31	18	15	99	237	385	572
	CDD10.0	1444	0	1	11	71	173	274	312	339	182	68	13	0
	CDD18.3	265	0	0	0	2	12	56	71	96	28	1	0	0
	CDH23.3	1679	0	0	0	9	64	323	415	700	167	1	0	0
	CDH26.7	381	0	0	0	0	4	61	90	192	34	0	0	0
Wind	WSAvg	3.2	3.0	3.6	3.8	4.1	3.4	3.2	2.9	2.6	3.1	3.0	3.1	3.0
Precipitation	PrecAvg	568	22	19	27	47	68	93	82	49	56	45	33	31
	PrecMax	770	44	41	55	96	110	213	127	92	147	90	71	64
	PrecMin	394	4	3	7	3	26	26	23	6	5	2	0	3
	PrecStd	107	12	10	15	22	25	41	32	25	30	25	16	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	14.1	19.7	24.6	27.1	30.1	30.8	32.4	29.8	22.6	18.0	10.4
		MCWB	6.4	9.0	10.9	14.2	16.4	21.0	20.9	20.2	18.2	14.6	11.7	7.2
	2%	DB	7.1	11.6	16.4	22.6	25.1	28.5	29.0	30.3	27.0	20.9	15.8	8.6
		MCWB	5.1	7.6	9.2	13.2	16.1	19.8	19.9	19.4	17.6	13.9	10.8	6.2
	5%	DB	5.4	9.8	13.9	20.5	23.6	26.7	27.3	28.8	25.0	19.4	14.0	7.1
		MCWB	3.8	6.5	7.7	12.2	15.4	19.3	19.4	18.9	16.3	13.0	10.2	5.0
	10%	DB	3.8	7.9	11.6	18.2	21.7	25.1	25.6	27.2	23.0	17.6	12.2	5.3
		MCWB	2.5	5.4	6.5	11.2	14.6	18.4	18.7	18.8	15.8	12.4	9.1	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	10.0	11.5	15.1	18.2	22.7	21.7	21.6	19.6	15.6	12.2	8.0
		MCDB	8.5	13.2	19.0	22.4	24.8	28.6	28.4	29.1	27.5	20.5	16.5	9.4
	2%	WB	5.5	8.0	9.6	14.0	17.2	21.0	20.8	20.7	18.3	14.6	11.3	6.4
		MCDB	7.1	10.9	15.6	20.9	23.3	26.7	27.1	27.6	25.1	19.6	14.9	8.5
	5%	WB	4.1	6.7	8.3	12.9	16.2	19.9	20.1	20.0	17.2	13.6	10.5	5.0
		MCDB	5.5	9.4	13.2	19.2	21.9	25.6	25.8	26.9	23.5	18.3	13.7	6.6
	10%	WB	2.6	5.5	7.0	11.8	15.3	18.9	19.4	19.3	16.4	12.7	9.5	4.0
		MCDB	3.6	7.8	10.7	17.3	20.4	24.2	24.5	25.7	21.9	16.9	12.3	5.3
Mean Daily Temperature Range	5% DB	MDBR	4.3	5.9	7.3	8.8	8.5	8.9	8.8	9.7	8.6	7.5	5.4	3.9
		MCDBR	5.6	8.6	10.4	11.5	10.7	10.6	10.7	11.4	11.1	10.6	7.5	6.0
		MCWBR	4.1	5.2	5.7	5.3	4.6	4.7	4.3	4.0	4.1	4.9	4.1	4.4
	5% WB	MCDBR	5.4	8.0	9.7	10.6	9.8	10.3	9.8	10.7	10.5	9.5	6.5	5.8
		MCWBR	4.2	5.2	5.4	5.1	4.6	4.9	4.4	4.1	4.3	4.6	3.8	4.3
Clear-Sky Solar Irradiance	taub	0.305	0.331	0.370	0.420	0.404	0.400	0.419	0.425	0.386	0.353	0.331	0.307	
	taud	2.351	2.315	2.268	2.155	2.260	2.298	2.249	2.230	2.310	2.393	2.406	2.377	
	Ebn at Noon	784	833	849	835	863	866	843	821	822	799	747	738	
	Edh at Noon	74	94	115	141	131	127	132	129	109	87	71	66	
All-Sky Solar Radiation	RadAvg	1.28	2.12	3.24	4.35	5.33	5.81	5.75	5.28	3.77	2.59	1.47	0.99	
	RadStd	0.17	0.25	0.40	0.44	0.49	0.49	0.49	0.48	0.42	0.31	0.19	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)		+0.74	N/A	N/A	+0.99	N/A	N/A	-152	-220	+92	+19

Nomenclature: See separate page